

DS_A1_Q4

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1 Introductions

A ROC curve is a tool to assessing the performance of classification models, the AUC, which also known as **Area Under the Curve** is the main standard to use in the assessment. It's calculated by setting different thresholds of the classifiers to calculate the **True Positive Rates(TPR)** and the **False Positive Rates(FPR)**, then plot them as points like (FPR, TPR), then make all the points form the ROC curve.

2 Calculation of ROC

The calculation of ROC is based on these steps,

2.1 Make predictions of the classifier

First step is to obtain the predicted probabilities from the classifier, which could be done by R coding. Instead of getting positive or negative labels, it just give the probabilities of each two classes.

2.2 Set different thresholds

The probability values should be compared to the thresholds. If the predicted value is greater or equal than the threshold, classify it as **positive** while **negative** in the contrast.

2.3 Calculate TPR and FPR

Table 1 could be obtained from predictions with different thresholds, then the TPR and FPR could be calculated by

$$\text{True Positive Rate(TPR)} = \frac{TP}{TP + FN}, \quad \text{False Positive Rate(FPR)} = \frac{FP}{FP + TN}$$

Table 1: Confusion Matrix Table.

	Truth Positive	Truth Negative
Predicted Positive	True Positive(TP)	False Positive(FP)
Predicted Negative	False Negative(FN)	True Negative(TN)

2.4 Plot ROC curve

With points of (FPR, TPR) under different thresholds, they could be connected into a curve, like Figure 1.

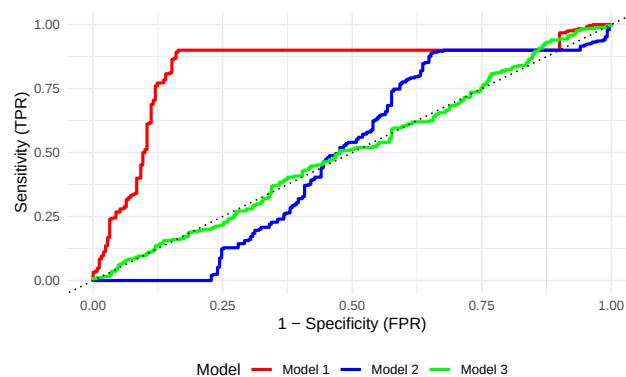


Figure 1: ROC curve

3 Interpretation of ROC

Mostly, the closer the ROC curve is to the top-left corner, the better the classifier. When it is closer to the diagonal line, it means the classifier's performance is closer to that of a random classifier. As Figure 1 shows, there are 3 different ROC curves together in the same plot, but they have different AUCs. The red curve (Model 1) has the greatest AUC which interprets that Model 1 has the best performance.

4 Advantages and Disadvantages of ROC Curves

4.1 Advantages

According to Tom Fawcett (Fawcett, 2006), ROC is a simple way to visualize and organize classifier performance. By plotting the ROC curve, the classification model's performance can be directly assessed. When visualizing becomes difficult, the AUC values can be compared instead.

Another advantage is that ROC is highly flexible for different datasets. It focuses on TPR and FPR, making it suitable for datasets with imbalanced class distribution.

4.2 Disadvantages

Using ROC curves requires knowing its properties and limitations, for example ROC is only focus on TPR and FPR, but could not reflect the precision and recall. While the outcomes showing with huge false positive or false negative values, the ROC curve could not be the only standard to assess the performance of classifiers. Besides, the ROC curve can only be used for binary classification models, when coming to circumstances of more than 2 response classes, ROC is not as useful.

5 References

Fawcett, T. (2006). An introduction to ROC analysis. Pattern Recognition Letters, [online] 27(8), pp.861–874. doi:<https://doi.org/10.1016/j.patrec.2005.10.010>.